

Machine Learning Assignment

1. Explain in your own words what a **coefficient represents in regression**. Also give one simple real-life example where a coefficient explains the relationship between two variables.

2. Consider the regression model:

$$Y = 8X + 15$$

Answer the following:

- Identify the **coefficient** and the **intercept**.
- Explain what the coefficient tells us about the relationship between **X** and **Y**.

3. A regression model used to predict marks is:

$$\text{Marks} = 6 \times \text{StudyHours} + 40$$

Explain the meaning of:

- The coefficient **6**
- The intercept **40**

Also explain what happens to the predicted marks if **study hours increase by 2 hours**.

4. A regression model is given as:

$$\text{Salary} = 12 \times \text{Experience} + 25$$

Calculate the predicted salary for the following experience values:

Experience (Years)	Predicted Salary
2 ?	
5 ?	
7 ?	

Show all calculations.

5. Consider the regression equation:

$$Y = -3X + 20$$

Answer the following:

- What does the **negative coefficient** indicate?
- What happens to **Y** when **X increases by 1**?
- Calculate the value of **Y** when **X = 4**.

6. Suppose a regression model used to predict house price is:

$$\text{Price} = 3000 \times \text{Size} + 40000 \times \text{Bedrooms} + 150000$$

Explain the meaning of the coefficients of:

- **Size**
- **Bedrooms**

According to the equation, which feature has a **larger impact on the house price**?

7. Write a Python program using **LinearRegression** to train a regression model using the dataset below.

Study Hours	Marks
1	50
2	55
3	60
4	65
5	70

Your program should:

- Train the regression model
- Print the **coefficient**
- Print the **intercept**

8. Using the regression model created in the previous question, write a Python program to **predict marks for 6 study hours** and display the predicted value.

9. Consider the dataset below:

StudyHours	SleepHours	Marks
1	7	50
2	6	55
3	7	60
4	6	65
5	8	70

Write a Python program to:

- Train a regression model using this dataset
- Print the **coefficients for both features**
- Print the **intercept**

10. Explain why **coefficients are important in regression models**. Also explain how coefficients help in understanding the **impact of input features on predictions**.